

A MULTILINEAR POLYNOMIAL WITH CONSTANTS WHOSE IMAGE ON $M_2(K)$ IS NOT A VECTOR SPACE

ABSTRACT. The concluding remarks of the e-print [arXiv:2312.13865v1](#) of Panja, Saini and Singh state, as Conjecture 2, that for every field K the image of every multilinear polynomial ω in the free product of $M(n, K)$ with the free algebra on $x_1, \dots, x_m$ is a vector space. Take $n = m = 2$ and $\omega(x_1, x_2) = x_1 e_{11} x_2$, a single monomial with one interior coefficient. Its image is exactly the set of matrices of rank at most one in $M_2(K)$, so it contains $\omega(I_2, I_2) = e_{11}$ and $\omega(e_{21}, e_{12}) = e_{22}$ but not their sum $e_{11} + e_{22} = I_2$; the printed statement is therefore false. Every entry of the witness is 0 or 1, so it serves over every field. We address only that sentence, in the version quoted; the published version of the source we were unable to read.

1. THE STATEMENT

Let K be a field and let $M_n(K)\langle x_1, \dots, x_m \rangle$ denote the free product of the matrix algebra $M_n(K)$ with the free algebra on $x_1, \dots, x_m$; its elements are the *polynomials with constants*, and each such ω induces a map $\tilde{\omega}: M_n(K)^m \rightarrow M_n(K)$ by evaluation.

Section 5 (*Concluding remarks*) of the e-print [1], on page 33, calls a polynomial *multilinear* “if it is linear in each variable”, offers

$$x_1 A_1 x_2 A_2 x_3 + x_2 A_4 x_1 x_3 A_5 - x_3 x_1 A_6 x_2, \quad A_i \in M(n, K),$$

as “a multilinear polynomial in 3 variables in algebra $M(n, K)$ ”, writes “Generalizing Theorem B, we present a generalized version of the L’vov-Kaplansky conjecture”, and states the following.

Conjecture 2 ([1], p. 33). Let K be a field. For any multilinear polynomial in $\omega \in M(n, K)\langle x_1, x_2, \dots, x_m \rangle$, the image $\tilde{\omega}(M(n, K))$ is a vector space.

The wording is as printed, including “polynomial in ω ”. By the displayed example, coefficients standing *between* variables are inside the quantifier, and no hypothesis of invertibility or nonvanishing on the A_i appears; the quantifier on K is “Let K be a field”, although [1] assumes algebraic closure for its theorems. The witness below is insensitive to the latter point.

That e-print is the only version on arXiv and carries the comment “Preliminary version; 34 pp;”. A published version appeared as *Mathematika* **71** (2025), no. 3, Paper e70031; the DOI returned HTTP 403 and we were

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unable to read it, so we make no claim about its wording. If the published concluding remarks restrict the positions of the coefficients, then what follows is a correction to the e-print rather than a counterexample to the published statement. We make no claim about the L'vov–Kaplansky conjecture, nor about the theorems of [1], nor about any statement other than the one displayed above.

2. THE COUNTEREXAMPLE

Take $n = 2$, $m = 2$, let K be any field, and let

$$\omega(x_1, x_2) = x_1 A x_2, \quad A = e_{11},$$

a single monomial with one interior coefficient, of the shape of the example quoted above. The object, and the two evaluation points that decide the matter, are these five matrices, written as lists of rows:

$$\begin{aligned} A &= [[1, 0], [0, 0]] \\ X_1 &= [[1, 0], [0, 1]] \\ Y_1 &= [[1, 0], [0, 1]] \\ X_2 &= [[0, 0], [1, 0]] \\ Y_2 &= [[0, 1], [0, 0]] \end{aligned}$$

All entries are 0 or 1, so the same five matrices are defined over every field through the unique ring map $\mathbb{Z} \rightarrow K$.

Theorem 1. *Let K be a field and $\omega(x_1, x_2) = x_1 e_{11} x_2$ in $M_2(K)\langle x_1, x_2 \rangle$. Then ω is multilinear and*

$$\tilde{\omega}(M_2(K)) = \{M \in M_2(K) : \text{rank } M \leq 1\}.$$

In particular $e_{11} = \omega(X_1, Y_1)$ and $e_{22} = \omega(X_2, Y_2)$ lie in the image while $e_{11} + e_{22} = I_2$ does not, so the image is not closed under addition and Conjecture 2 of [1] is false.

Proof. Multilinearity is immediate: ω is a single monomial containing each of x_1, x_2 exactly once, and $(X, Y) \mapsto XAY$ is K -bilinear.

Write $e_1 = (1, 0)^\top$, so that $A = e_1 e_1^\top$. Then for all $X, Y \in M_2(K)$

$$(1) \quad XAY = (X e_1)(e_1^\top Y) = u v^\top,$$

where u is the first column of X and $v^\top$ is the first row of Y . Every such product is an outer product and therefore has rank at most 1. Conversely a matrix of rank at most 1 is $uv^\top$ for some $u, v \in K^2$ (rank 0 being $u = 0$), and choosing X with first column u and Y with first row $v^\top$ realises it as XAY by (1). This proves the displayed equality.

For the two witnesses, $X_1 = Y_1 = I_2$ gives $\omega(I_2, I_2) = A = e_{11}$, and $X_2 = e_{21}$, $Y_2 = e_{12}$ give $\omega(e_{21}, e_{12}) = e_{21} e_{11} e_{12} = e_{21} e_{12} = e_{22}$. Both have rank 1. Their sum is I_2 , and $\det I_2 = 1 \neq 0$ in every field, so $\text{rank}(e_{11} + e_{22}) = 2 > 1$ and $I_2 \notin \tilde{\omega}(M_2(K))$. $\square$

Remark. The image contains $0 = \omega(0, 0)$ and is closed under scalar multiplication, since $\lambda(XAY) = (\lambda X)AY$; of the vector space axioms only additivity fails.

3. WHAT WAS CHECKED

The proof above is by hand and needs no machine. The accompanying program `verify.py` uses the Python standard library only, decides nothing in floating point, and uses no randomness. It parses the five matrices out of the verbatim display in Section 2 rather than carrying its own copy, and from them re-derives: the two evaluations and all four ranks, over $\mathbb{Q}$ and over $\mathbb{F}_p$ for $p \leq 11$; the factorization (1) on every ordered pair of matrix units of M_n for $n \leq 4$ over $\mathbb{Z}$ and on all 262,144 ordered pairs of $M_3(\mathbb{F}_2)$; multilinearity by exhaustion; and the set equality of Theorem 1 over $\mathbb{F}_2$ and $\mathbb{F}_3$. The recorded run `verify.output.txt` reports **VERDICT: ALL 89 CHECKS PASS** and exit status 0; the checks it reports beyond those just listed concern statements this note does not make. No enumeration runs over an infinite field, and none is needed: the proof of Theorem 1 is uniform in K . The program cannot reach the published version of [1], so the qualification recorded in Section 1 is not machine checked.

REFERENCES

- [1] S. Panja, P. Saini, and A. Singh, *Images of polynomial maps with constants*, e-print arXiv:2312.13865v1, 2023, the version quoted throughout this note. A published version appeared as *Mathematika* **71** (2025), no. 3, Paper e70031, doi:10.1112/mtk.70031; we were unable to read it.